Development of an MEG Compatible Musical Instrument *

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***Abstract*— Several musical instruments for use with functional Magnetic Resonance Imaging have been described, but no multi-note instrument has been documented that is explicitly compatible with magnetoencephalography (MEG). Such an instrument would allow deeper exploration of brain-music interaction on short timescales, with specific applications to improvements in music therapy and diagnosis of autism spectrum and other disorders. This paper describes a 25-key all-polymer keyboard, connected by fiber optic cables to an electronics box that resides outside of the magenetically shielded room or zone of potential interference. A three-state optical encoder indicates velocity and key state up or down. The device can successfuly play simple melodies (such as "Happy Birthday") as well as chords. Different speeds of key press produce scaled volumes. The device will be tested using MEG to verify non-interference and for preleminary evaluation of the any muscular interference caused by playing the instrument.**

I. INTRODUCTION

Studying the real-time neurological activity that occurs during music performance not only promises insight into how the brain processes music, but may also help understand and optimize music therapies for various disorders, such as autism spectrum disorders (ASD) [4]. In particular, delays in a 100ms latency in response to stimulus has been suggested as a biomarker for ASD [1], and also appears to change with musical training [3]. While musical instruments exist for functional magnetic resonance imaging (fMRI), the timescales required to answer these questions are too short for fMRI. An MEG compatible musical instrument would permit researchers to use the MEG's fine temporal resolution to precisely measure time differences such as these in neural signals

While the only documented musical instruments for MEG-based studies to date have been mono-tonal [14], [15], there is a wide array of those designed for fMRI use. Approaches have included passively listening to music [7], playing a mute keyboard [7], [8], and playing an MRI-compatible instrument that provides real-time auditory feedback[9], [10]. Several of these MRI-compatible instruments have been described in the literature, including a keyboard [5] and cello[12]. Of the instruments, only [5] describes the design of a multi-tonal instrument compatible with the spatial and movement constraints of MEG. Even so, the design is not explicitly MEG compatible, and offers considerable opportunity for improvement.

This paper describes the design and preliminary testing of an MEG compatible 25-key full-size piano keyboard. The instrument outputs MIDI with velocity-controlled volume modulation and a trigger signal to synchronize the key presses, and is capable of playing multiple notes simultaneously. In conjunction with a speaker or MEG compatible earphones, the instrument allows researchers to study real-time feedback loops during musical performance on a timescale not possible with fMRI.

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II. BACKGROUND

Many studies of music-brain interaction have used fMRI, ranging from the long-term effects of musical study [6] , passively listening to music [7], playing a mute keyboard [7], [8], and playing an MRI-compatible instrument that provides real-time auditory feedback [9], [10], [2]. For understanding the real-time effects of music performance, only instruments providing auditory feedback can be ecologically valid.

The designs of several of these musical instruments have been described for fMRI use, including a Ballagumi, a flexible silicone interface embedded with fiber optic sensors [11], a cello [12], and a piano keyboard [11] . The Ballagumi provides a novel interface, which allows studies of motor learning in connection to music but limits its use in music therapy applications and studies of practicing musicians. A cello allows study of pitch modulation, but its size and the movement required of the player near the head and in the shoulder limit its MEG applications. The keyboard described in [11] while fiber-optically sensed, is not explicitly MEG compatible and describes several limitations in reliability and robustness in design which this paper improves upon.

A MIDI instrument [15] has been described for MEG use, but as it consists of a single piezo button, it is not capable of playing discrete notes with an interface familiar to musicians and non-musicians alike. Furthermore, while the device did not produce obvious artifacts, the use of electric potential as the sensing method requires investigation into the interfence that might come from scaling to more buttons.

III. DESIGN

A. Functional Requirements

Several high-level requirements guided the design, ensuring that it would allow researchers to collect meaningful data.

1) The device itself must not significantly distort the MEG signal
2) Use of the device must not significantly distort the MEG signal
3) The device must be musical-instrument-like
    - User must be able to control pitch and timing of multiple notes
    - Latency from key-press to audio must be less than can be perceived, approx. 25 ms

- The device should have an interface familiar to professional musicians and to non-musician subjects.

4) The device must be able to produce music
   - The device must play at least eight notes from a standard western musical scale
   - The device must be able to play more than three notes simultaneously
5) The device must indicate or record timing and identity of notes played.
6) The device must fit physically within the constraints of an MEG.

The majority of these requirements have clear criteria against which to evaluate the prototype. The threshold at which user movement causes unacceptable interference is somewhat subjective, and highly dependent on analysis methods and artifact-reduction strategies. Further characterization of this threshold is an opportunity for future study, and the described device can enable such investigation.

B. Design Approach

In developing concepts to satisfy the above requirements, several sub-categories emerged: format of the instrument, method of sensing input, method of transmitting input out of the magnetic shield, and method of processing input into audio and a time marker for imaging purposes. In one strategy, an exsisting instrument produces the audio and a laser or acoustic microphone with known latencies records the event timing. In another strategy, the input event is detected electrically or optically and then synthesized to produce the audio.

Format: A standard piano keyboard was chosen for its familiarity to a wide range of subjects, and low barrier to musical expression relative to other instruments. Additionally, piano playing naturally concentrates movement at the hands, limiting potential for interference-causing muscle activation near the MEG sensors.

Sensing: Transmissive optical sensing of an encoded signal offers robust preservation of information over an unencoded analog signal, as well as simple and reliable alignment of the fibers compared to reflective sensing.

Transmission: Plastic optical fiber allows transmission of an encoded optical signal outside of the magnetically shielded room without the risk of picking up RF interference that running an electrical wire might bring. Polymer construction and no electric current further ensure compatibility.

Processing: The availability of MIDI synthesizers favored the approach of sensing an input event and producing audio with measurable latency, rather than back-calculating latencies to determine from an audio event when a note was played. In this design two comparators feed state signals to a microprocessor, which calculates velocity and onset note timing and outputs a MIDI signal to a synthesizer and a trigger to the MEG data acquisition system.

C. System Overview

The proof-of-principle prototype is depicted in Fig. 1. When a key is pressed, a simple three-state encoder passes from transparent to opaque to semi-transparent across two fiber optic cables. This modified light signal is compared to two thresholds to determine both time (and hence velocity), and state (up or down). An MSP 432 microcontroller unit (MCU) interprets these signals to output send a note-on or note-off signal to the synthesizer over MIDI, and a trigger signal to the MEG data acquisition interface. Audio from the sythesizer is fed to the subject via a speaker or MEG-compatible headphones.

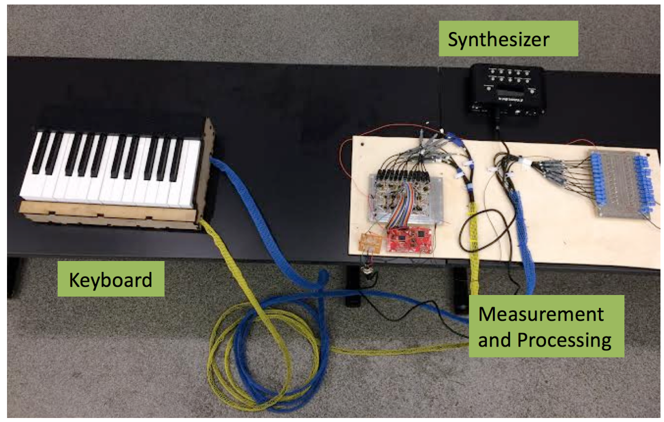

Fig. 1. System Overview

D. Sensing Mechanics

A 3-state optical encoder, which is placed in between transmitting and receiving optical fibers is used to turn the motion of the piano key into varying light intensity. The encoder (Fig. 2) consists of three bands with different transparencies printed on standard transparency film. The transparent state indicates a raised key, while the semi-tansparent state (set to 33% opacity in inkscape) indicates a depressed key. With a constant 4mm opaque band (100% opacity), key velocity is directly proportional to the time between transparent and semi-transparent signals. Figure 2 further shows optical intensity over a key press (transparent $\rightarrow$ opaque $\rightarrow$ semi-transparent) and release (semi-transparent $\rightarrow$ opaque $\rightarrow$ transparent).

E. Electronics

The circuits shown in Fig. 3 convert light intensity into voltage signals and determine key state and press velocity. A transmitting LED (IF E97, i-fiberoptics.com) outputs a constant signal at 660nm; phototransistor (IF D92, i-fiberoptics.com) receives the encoded signal. A transimpedance amplifier (LM6132, digikey.com) outputs an analog voltage V_{out} which follows the light intensity signal, determined by

$$V_{\text{out}} = i/R_f \tag{1}$$

Two comparators (TLV3702, digikey.com) at thresholds A and B determine key state: threshold voltage A is set to

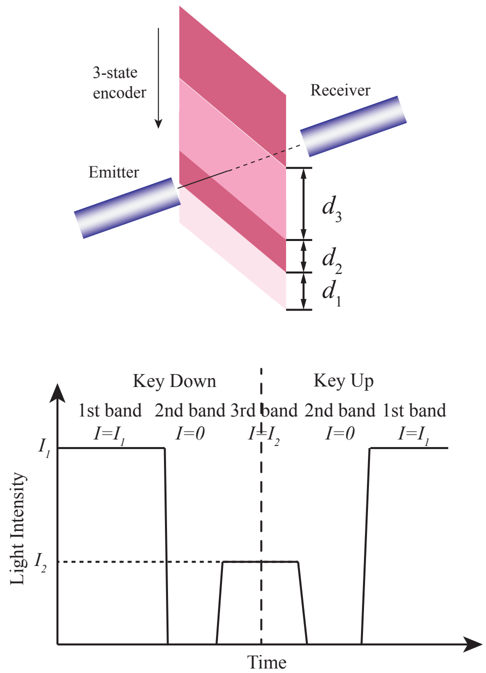

Fig. 2. Three-State Encoder and Optical Output

be lower than V_1, while threshold voltage B is set to be lower than V_2 and higher than V_1. The result is two signals, one (B) indicating whether the optical fiber is aligned with the transparent band of the encoder, the other (A) indicating whether the optical fiber is aligned with either the transparent band or the semi-transparent band of the encoder. If A is high and B is low, the key is down; if A and B are high, the key is up.

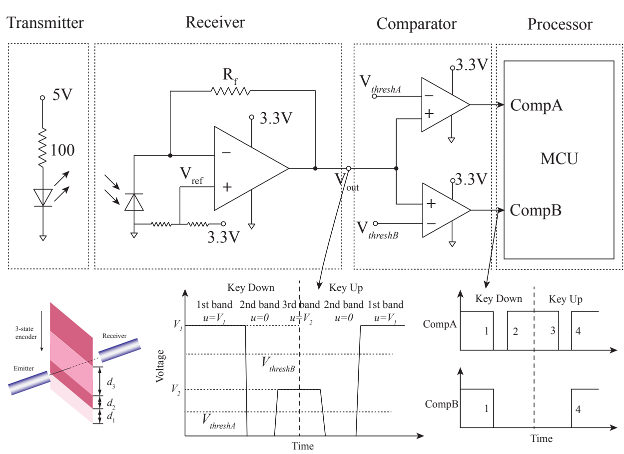

Fig. 3. Circuit Schematics for Sensing Electronics

F. Processing

The digital signals A and B from the comparators are fed to interrupt pins on the MCU. Signal A triggers an interrupt in the MCU to calculate the velocity of the key press, while signal B is polled in the main code to check whether a key press has been completed and the key as risen to its resting position. For the purposes of clarification and logic flow, 'flag' and 'MIDI' are depicted as state variables and are initialized to 0. When Interrupt A triggers an interrupt on the MCU, and enters the interrupt service routine (ISR), there is a check on the flag variable. If the flag is 0 (cleared), (flag = 0 corresponds to transition 1 and 3), then proceed with check on MIDI value. If the MIDI flag is 0 (cleared), then grab the value of a timer that is running in the background and store in a variable, switch interrupts to be serviced on rising edges and set flag = 1, and exit ISR. On transition 2, enter the ISR again (since interrupts are now triggered on rising edges) and flag = 1. Therefore proceed with retrieving a second timer value, and using it along with the first timer value, calculate the speed of a key press, send the MIDI note ON message with the corresponding volume information, set flag = 0, set MIDI = 1, and switch transition trigger to falling edge once again. Now enter on transition 3, flag = 0, but MIDI = 1, which means do nothing. This ensures that transition 4 does not trigger an interrupt (for edge case reasons). In Main loop, there is constantly a check running for a note off message. This involves checking if a MIDI note ON message has been sent out (MIDI =1), and that Signal B is now High. Once this event occurs (which it does after transition 4), the main loop code runs this for every key and sends out relevant MIDI note OFF messages, as well as resetting MIDI = 0.

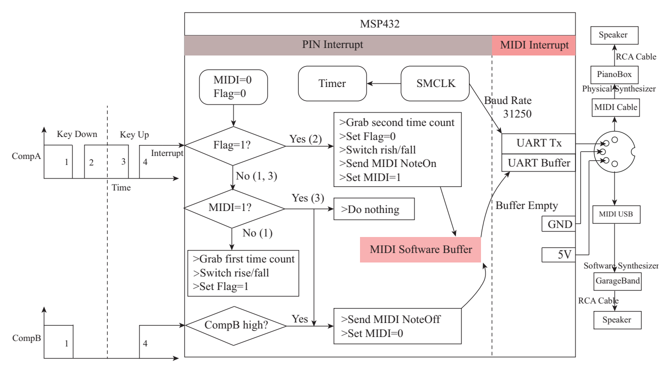

Fig. 4. Processing Logic Flow

G. Sound Module

In order for the subject to hear real music notes from pressing keys, information about key note and key velocity has to be synthesized to generate sound. Musical Instrument Digital Interface (MIDI) is the language used to turn digital electrical signals into music as a series of bytes transferred through UART serial communication. In this case, the MCU converts key note and velocity information into machine-recognizable bytes and sends them out through MIDI-compatible hardware to the speaker that plays the desired musical notes.

The MCU sends two kinds of MIDI messages; NoteOn, which turns on a musical note with the volume proportional to the velocity, and other called NoteOff, which turns off the musical note. Both messages contain 3 bytes: the first byte indicates whether it is a NoteOn or NoteOff message, and the second and third bytes indicate the note and volume/velocity. Figure 4 shows the overall MIDI schematic. To ensure compatibility with the MIDI hardware used to generate the musical notes, the UART baud rate is configured to be 31250bps. After identifying the keys that are being pressed and their corresponding velocities, MIDI messages are generated and stored in a large data array that acts as a software buffer. MIDI messages are sent through UART transmission port during the ISR, where the MCU is constantly checking if the UART transmission buffer contains data. If the buffer is empty, the MCU will enter the interrupt loop and pass messages from the large data array to the UART buffer. In this case, the UART transmission is not in the main key ISR, so does not slow down the pin interrupt loop, which makes it possible both to play chords and to play notes quickly.

A standard MIDI 5-PIN connector is used to communicate between the MCU and either a physical synthesizer, such as the PianoBox USB (Miditech), or a software synthesizer, such as GarageBand (Apple) can be used.

H. Packaging and Key Mechanics

To preserve the dimensions and texture of a familiar piano keyboard, the design incorporated replacement keys for a YAMAHA electric keyboard (K64UC Style 64U, purchased from syntaur.com). For key return, ULTEM springs were chosen to approximate a 60g touchweight (measured at note onset).

Figure 5 shows the encoder block assembly. The encoders are adhered to replaceable clips that are press-fit into existing features on the keys. The fibers are held in alignment by modified "connectorless" mating sleeves (IF CS2, i-fiberoptics.com), which are threaded into an encoder block. Fingers on the encoder block maintain appropriate key spacing, and a narrow slit ensures the encoder is perpendicular to the fiber ends. The encoder block further serves to limit key motion to 1cm.

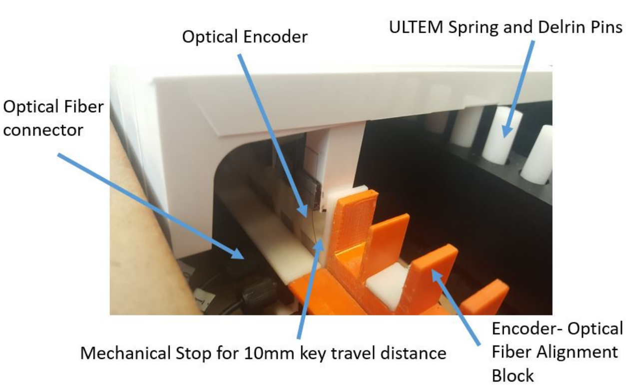

Fig. 5. A closer view of the optical encoder module

The prototype packaging (fig. 6) consists of a laser-cut enclosure fastened to the internal structural components by nylon machine screws. The internal structures are located on a base plate by plastic locating pins to preserve alignment after re-assembly. The space in front of the keys accomodates the 30mm recommended minimum bend radius. For the purposes of testing and demonstration, the electronics were mounted exposed on a plywood board with adequate strain relief; the electronics will be packaged in an enclosure with panel mounted Versatile Link fiber optic connections.

Fig. 6. Final Piano Keyboard Assembly

IV. TESTING

The latency of the instrument between comparator signal and audio output was 5ms, measured by oscilloscope. Even if the audio source is 5m away, with the speed of sound in air at 343 m/s, the additional lag is 15ms; assuming the delay from the analog electronics is insignificant, the total latency is reliably beneath the 25ms threshold for human perception.

The following protocol (approved by the MIT Committee on the Use of Humans as Experimental Subjects) is planned to evaluate device and subject interference:

Moving the keyboard assembly under the sensor dome and checking for any signal disturbance will confirm its compatibility.

To determine movement artifacts, the subject will be seated next to the MEG, with their right shoulder and arm reaching inside to the keyboard, which will be placed in front of the empty chair. This will establish a baseline signal with no brain in the sensor dome.

The subject will then play single notes, chromatic cales, and then progressive chords, each requiring muscle activation in the shoulder the MEG signal will be recorded, still with an empty dome. These results will offer preliminary clues to the extent of movement allowable.

The timing of the musical notes played by the subjects will be recorded so that the timestamp of the music being played with the timestamp of the MEG data recorded can be aligned. This will allow for signal averaging over a narrow window around the muscular event.

V. CONCLUSIONS

The musical instrument described offers researchers an opportunity to probe the neural patterns associated with musical performance at a fine temporal resolution. The proof-of-concept prototype demonstrated the ability to play an octave of western scale and multiple simultaneous notes while satisfying other requirements; MEG testing will confirm material compatibility. Future work on this instrument includes a robust electronics enclosure and circuit board design and fabrication, in addition to characterization of the motion artifacts generated from playing a keyboard during MEG.

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